

System Development Life cycle (SDLC)

CHAPTER TWO



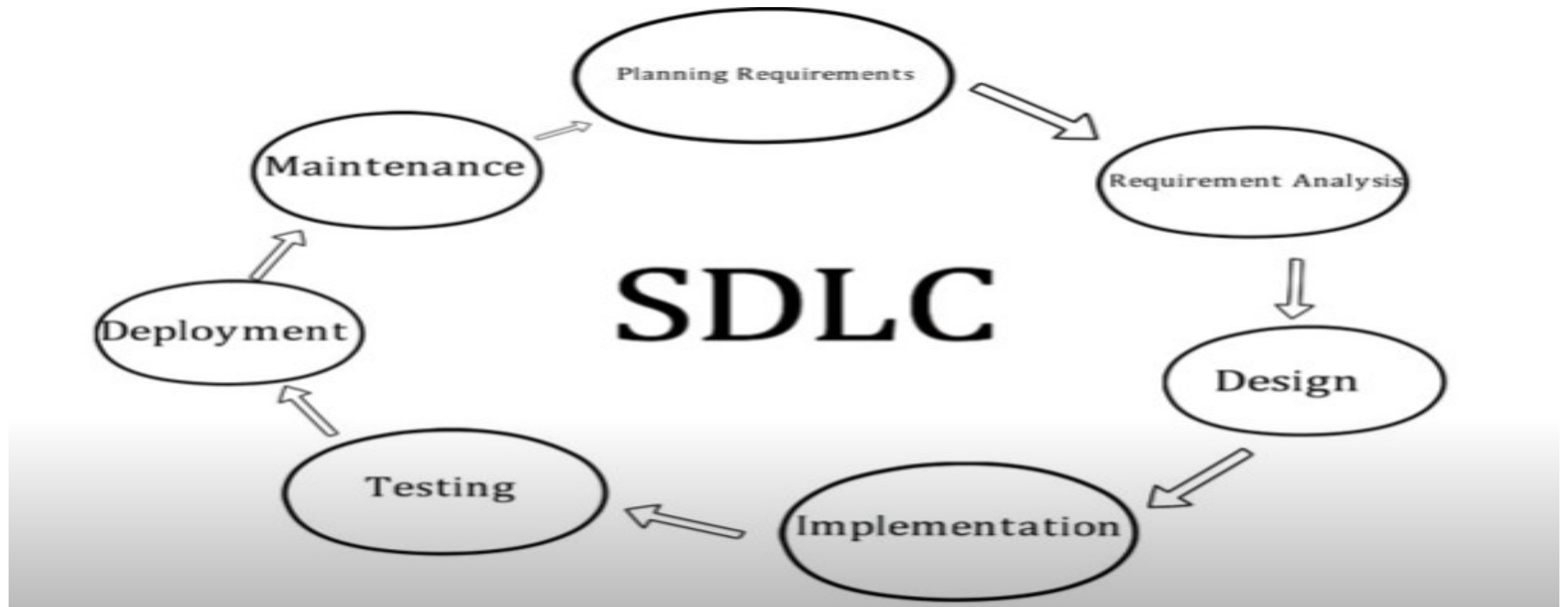
SDLC

- It is a systematic approach which explicitly breaks down the work into phases that are required to implement either new or modified Information System.
- SDLC is used by analysts to develop an information system.

SDLC includes the following activities :

- requirements
- design
- implementation
- testing
- deployment
- maintenance

Phases in SDLC



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- Examples of project selection criteria:
 - What is the **payback or break-even point**?
 - What is the **impact** to the organization's **growth** (e.g., customer volumes)?
 - Does the project **contribute to innovation**?
 - How much **risk** is involved?
 - Are there **sufficient resources** in terms of time, budget, infrastructure, and people with relevant expertise?
 - What is the **cost/benefit** ratio?

Phases of the SDLC

1. Planning /Feasibility Study

- This is **the first phase** in the systems development process.
- It identifies if there is the **need** for a new system to **achieve** a business's strategic **objectives**.
- This is a preliminary plan (or a feasibility study) for a company's business initiative **to acquire the resources** to build on an infrastructure to **modify or improve** a service.

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- The purpose of this step is to find out the scope of the problem and determine solutions.
- Resources, costs, time, benefits and other items should be considered at this stage.

Phases of the SDLC

□ *At this phase:*

- Define the **problem** and **scope** of existing system.
- Overview/Define the **new system** and determine its **objectives**.
- Confirm **project feasibility** and produce the **project Schedule**.
- During this phase, **threats**, **constraints**, **integration** and **security** of system are also considered.
- A **feasibility report** for the entire project is created at the end of this phase.

Feasibility Study/ Types

➤ Economic Feasibility

- It is evaluating the effectiveness of candidate system by using **cost/benefit** analysis method.
- It **demonstrates the net benefit** from the candidate system in terms of benefits and costs to the organization.
- The **main aim** of Economic Feasibility Analysis (EFS) is to **estimate the economic requirements of candidate system** before **investments funds** are committed to proposal.
- It prefers the **alternative** which will **maximize** the **net worth** of organization by earliest and highest **return of funds** along with **lowest level of risk** involved in developing the candidate system.

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➤ Technical Feasibility

- It investigates the **technical feasibility** of each implementation alternative.
- It analyzes and determines whether **the solution** can be supported by **existing technology** or not.
- The analyst determines whether current **technical resources be upgraded** or **added** to fulfill the **new requirements**.
- It **ensures** that the **candidate system** provides **appropriate responses** to what extent it can support the technical enhancement.

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➤ Operational Feasibility

- It determines whether the system is **operating effectively** after implementation.
- It ensures that **the management should support** the proposed system and its' work in the current organizational environment.
- It analyzes whether the users **will be affected** and they **accept the modified** or **new business methods** that affect the possible system benefits.
- It also ensures that the **computer resources and network architecture** of candidate system are workable.

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➤ Behavioral Feasibility

- It **evaluates** and estimates the **user attitude or behavior** towards the development of **new system**.
- It helps in determining if the system requires **special effort to educate, retrain, transfer, and changes in employee's job status** on new ways of conducting business.

➤ Schedule Feasibility

- It ensures that the project **should be completed within given time constraint or schedule**.
- It also verifies and validates **whether the deadlines of project are reasonable or not**.

Phases of the SDLC

2. Requirements Analysis

- This is where **teams** consider and **define** the functional **requirements** of the project or solution.
- It is also where **system analysis** takes place or analyzing the needs of the end users to **ensure the new system can meet their expectations**.
- Systems analysis is vital in **determining what a business's needs** are, as well as **how they can be met**, **who will be responsible for individual pieces of the project**, and **what sort of timeline should be expected**.

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Steps for preparing SRS Documentation

1. Gather, analyze, and validate the information.
 2. Define the requirements and prototypes for new system.
 3. Evaluate the alternatives and prioritize the requirements.
 4. Examine the information needs of end-user and enhances the system goal.
 5. A Software Requirement Specification (SRS) document, which specifies the software, hardware, functional, and network requirements of the system is prepared at the end of this phase.
- E.g.: Define User Registration to have UN input field, Password field, checkbox-to accept terms and conditions, submit button, save user to Database

Discussion time!

- Give title for the above example and write the functional requirements that will be covered in the system to be produced.
- List some functional requirements for your project title.

Phases of SDLC

3. System Design

- Includes the design of application, network, databases, user interfaces, and system interfaces.
- Transform the SRS document into logical structure, which contains detailed and complete set of specifications that can be implemented in a programming language.
- Create a contingency, training, maintenance, and operation plan.
- Review the proposed design.

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- **Ensure** that the **final design** must meet the requirements stated in **SRS** document.
- Finally, **prepare** a **design document** which will be used **during next phases**.
- **E.g.:** design includes UI layout, business logic, color scheme, Language to use, frameworks, server design, DB Relationships, ARCH the App, Mobile Aspect, Supported Browsers .

Phases of SDLC

4. Implementation/Coding

- Implement the design into source code through coding.
- Combine all the modules together into training environment that detects errors and defects.
- Integrate the information system into its environment and install the new system.
- E.g.: the operations team setup physical HW, developers start to code, designers Plan the UI, testers start building the test plan and test case,

Phases of SDLC

5. Testing

- Most **important phase** after the developer has finished **coding** the App.
- During the testing phase, developers find out **whether their code and programming work according to customer requirements**.
- Before testing can begin, the project team develops a **test plan**.
- The test plan **includes** the **types** of testing you'll be using, **resources** for testing, **how** the software will be tested, **who should be the testers** during each phase, and **test scripts**, which are instructions each tester uses to test the software.
- Types of testing during the test phase, including **quality assurance testing (QA)**, **system integration testing (SIT)**, and **user acceptance testing (UAT)**.

Phases of SDLC

6. Deployment

- This phase puts the product into production.
- After the project team tests the product and the product passes each testing phase, the product is ready to go live.
- This means that the product is ready to be used in a real environment by all end users of the product.

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- There are various phases of the deployment process the project team must follow the phases include
 - ✓ deployment preparation and procedures,
 - ✓ product deployment
 - ✓ transferring ownership of the product
 - ✓ closing the deployment phase.

E.g.: The operational team make the environment ready for production: installing HW, server, setting up the URL link, DB, finally host the app.

Phases of SDLC

7. Maintenance/Support/Operations

- This step is when end users can adjust the system, if they wish, [to boost performance](#), add new capabilities or meet additional user requirements.
- Include all the activities such as phone support or physical on-site support for users that is required once the system is installing.
- Implement the changes that software might undergo over a period of time, or implement any new requirements after the software is deployed at the customer location.

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- It also includes handling the **remaining errors** and **resolve any issues** that may exist in the system even **after the testing** phase.
- Maintenance and support may be needed for a **longer time for large systems** and **for a short time for smaller systems**.
 - **E.g..:** Since so many users logging in, registering in to the App. Thus, to increase performance we bring larger server, larger DB.

Discussion time!

Give some examples of reasons for a given system to put the maintenance phase into practice!

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